Monte Carlo simulation

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Summary

Estimating an expectation by simulating years, and how far off the estimate is.

- An expected annual damage is an integral that usually has no closed form, so we average simulated years instead.
- The law of large numbers says the average converges, given independent draws, identical distribution, and a finite mean, and says nothing about how fast.
- The estimator is unbiased: on average it lands on the right answer for any n.
- Its variance falls like $1/n$, so the error falls like $1/\sqrt{n}$ and halving the error costs four times the draws.
- Independence is what licenses splitting the variance of the sum, so draws that covary make the estimator noisier than n independent ones.
- The relative error depends on the coefficient of variation of the quantity being averaged, which is a property of that quantity and not of how the simulation is run.

Key results.

$$\theta = E[g(X)], \qquad \hat{\theta}_n = \frac{1}{n}\sum_{i=1}^{n} g(X_i), \qquad E[\hat{\theta}_n] = \theta$$

$$\mathrm{SE}(\hat{\theta}_n) = \frac{\mathrm{sd}(g(X))}{\sqrt{n}}, \qquad \frac{\mathrm{SE}(\hat{\theta}_n)}{\theta} = \frac{\mathrm{CV}}{\sqrt{n}}$$

Lab 1 breaks the finite-mean assumption and shows what happens, and has you find the coefficient of variation for several distributions to see which converge.

Bibliography